

Leaves of Large Type in the Numerical-Semigroup Tree

A genus-34 counterexample and a sharp asymptotic order

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20 September 2026

Abstract

We give an explicit counterexample to Conjecture 4.4 of Chappelon, Ramírez Alfonsín, and Stamate on the type of leaves of the numerical-semigroup tree. The numerical semigroup

$$S = \{0, 26, 27, 29, 31, 32\} \cup [39, 50]_{\mathbb{Z}} \cup [52, \infty)_{\mathbb{Z}}$$

has genus 34, type 17, and no minimal generator above its Frobenius number 51; the conjectured bound is 16. Its validity is proved directly, without relying on a search program. We then turn restricted additive bases into infinite families of leaves with explicitly determined pseudo-Frobenius sets. One elementary family has genus $11q + 1$ and type $6q - 1$ and contradicts the conjecture for every $q \geq 3$.

More generally, let $L(g)$ be the maximum type of a leaf of genus g . A sparse basis construction, a genus-padding operation, and a counting argument in Apéry sets prove

$$L(g) = \frac{2}{3}g - \Theta(\sqrt{g}).$$

In particular, the extremal type-to-genus ratio tends to $2/3$, not $1/2$. Explicit two-sided bounds hold at every genus $g \geq 7$. The archive includes an exact-arithmetic verifier, independently reconstructed Apéry certificates, finite enumeration data, the random area-selection record, and optional search code. The mathematical arguments are self-contained; the computational tests are additional checks, not substitutes for the infinite proofs.

Status. This note disproves the stated conjecture and proves the displayed asymptotic order. It does not determine the exact function $L(g)$ or its sharp square-root coefficient. The retrieved version of the source paper still states the conjecture; a targeted search did not locate a later resolution. Priority and the proofs have not been independently refereed.

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1 The problem and the outcome

The area was selected before choosing the problem, using a pseudorandom generator seeded from operating-system randomness. A uniform draw from 24 areas selected *semigroup theory*. The seed and the full sampling frame are recorded in section B and in the accompanying JSON file.

The target is Conjecture 4.4 of Chappelon, Ramírez Alfonsín, and Stamate [1]. For $g = 4q + r \geq 2$, with $0 \leq r < 4$, put

$$\beta(g) = \begin{cases} 2q - 1, & r = 0, 1, \\ 2q, & r = 2, \\ 2q + 1, & r = 3. \end{cases} \quad (1.1)$$

The conjecture asserts that every leaf S of genus g satisfies $t(S) \leq \beta(g)$. Notice the strict inequality

$$\beta(g) < \frac{g}{2}. \quad (1.2)$$

Thus a leaf with $2t(S) \geq g(S)$ necessarily contradicts the conjecture.

The counterexample proved in section 3 has $(g, t) = (34, 17)$. The source paper tabulates leaf counts through genus 33 [1, Table 5], so the example occurs immediately beyond that reported range. Our own exhaustive enumeration stops at genus 20; we do not claim an independent exhaustive proof that genus 34 is globally minimal.

The counterexample suggests an extremal problem that admits a substantially stronger answer. In this note, define

$$L(g) = \max\{t(S) : S \text{ is a leaf and } g(S) = g\}. \quad (1.3)$$

The symbol denotes a *maximum type*, not a number of leaves. The sets in (1.3) are nonempty for every $g \geq 7$, as our constructions show, and they are finite. Our main quantitative result is

$$\frac{2g}{3} - \sqrt{24g - 87} - 2 \leq L(g) \leq \left\lfloor \frac{6g + 2 - \sqrt{24g - 23}}{9} \right\rfloor, \quad g \geq 7. \quad (1.4)$$

Both errors from $2g/3$ have square-root order. The proof has two independent parts: a construction for the lower bound and an Apéry-set counting argument for the upper bound. Neither uses an unproved classification of numerical semigroups or an assumed optimality result for additive bases.

2 Definitions and finite certificates

We use $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and $[a, b]_{\mathbb{Z}} = \{n \in \mathbb{Z} : a \leq n \leq b\}$. The notation $A + B$ means $\{a + b : a \in A, b \in B\}$.

Definition 2.1. A *numerical semigroup* is an additive submonoid $S \subseteq \mathbb{N}_0$ with finite complement. Except when describing the root of the semigroup tree, we assume $S \neq \mathbb{N}_0$. Its gap set, genus, Frobenius number, conductor, and multiplicity are, respectively,

$$\begin{aligned} G(S) &= \mathbb{N}_0 \setminus S, & g(S) &= |G(S)|, & F(S) &= \max G(S), \\ c(S) &= F(S) + 1, & m(S) &= \min(S \setminus \{0\}). \end{aligned}$$

A positive element is *primitive* if it is not a sum of two positive elements of S . The primitive elements form the minimal generating set; its cardinality $e(S)$ is the embedding dimension.

To see why the primitive elements generate, repeatedly split a nonprimitive positive element into two smaller positive elements. This process terminates because positive integers decrease. Conversely, a primitive element must belong to every generating set. A primitive element cannot exceed $F + m$: if $n \geq F + m + 1$, then $n = m + (n - m)$ and $n - m \geq F + 1$ belongs to S .

Definition 2.2. The *numerical-semigroup tree* has root $\mathbb{N}_0$. The parent of a nonroot semigroup S is $S \cup \{F(S)\}$. Equivalently, its children are obtained by deleting one primitive element larger than $F(S)$. A semigroup is a *leaf* when it has no children, or, equivalently, when every primitive element is at most its Frobenius number.

For completeness, adjoining F preserves closure: every sum involving F and a positive semigroup element is larger than F . Deleting a primitive $x > F$ also preserves closure, since a sum of two remaining positive elements cannot be x . The new largest gap is x , so these operations give exactly the stated parent-child relation.

Definition 2.3. A *pseudo-Frobenius number* is a gap $p \in G(S)$ such that

$$p + s \in S \quad \text{for every } s \in S \setminus \{0\}.$$

Write $\text{PF}(S)$ for this set and $t(S) = |\text{PF}(S)|$ for the *type*.

The largest gap F always belongs to $\text{PF}(S)$, so $t \geq 1$. To verify that a gap is pseudo-Frobenius, it suffices to test addition by minimal generators. Indeed, after the first generator is added the sum belongs to S , and further additions preserve membership. Alternatively, testing all positive elements at most F is finite and sufficient, because larger additions already exceed the Frobenius number.

These observations give small certificates for statements that initially quantify over an infinite semigroup. Closure only needs to be checked below the conductor. The leaf condition only needs decompositions of the interval $[F + 1, F + m]_{\mathbb{Z}}$. A non-pseudo-Frobenius gap needs just one positive element whose addition leaves another gap.

3 A counterexample of genus 34

Theorem 3.1. *Let*

$$A = \{26, 27, 29, 31, 32\} \cup [39, 50]_{\mathbb{Z}}, \quad S = \{0\} \cup A \cup [52, \infty)_{\mathbb{Z}}. \quad (3.1)$$

Then S is a leaf with

$$(m, F, c, e, g, t) = (26, 51, 52, 17, 34, 17).$$

Its minimal generators are exactly A , and

$$\begin{aligned} \text{PF}(S) = & \{13, 14, 15, 16, 17, 18, 21, 23\} \\ & \cup \{28, 30, 33, 34, 35, 36, 37, 38, 51\}. \end{aligned} \quad (3.2)$$

Consequently, $t(S) = 17 > \beta(34) = 16$.

Proof. Every positive element of S is at least 26. The sum of two positive elements is therefore at least 52 and belongs to S . This proves closure; the full tail proves cofiniteness. The multiplicity is 26, and the largest gap is 51.

There are five elements in $\{26, 27, 29, 31, 32\}$ and twelve in $[39, 50]_{\mathbb{Z}}$. Hence A has 17 elements and

$$g = 51 - |A| = 34.$$

Every element of A is primitive, because it is smaller than $2m = 52$. We show that there are no others.

Put $D = \{0, 1, 3, 5, 6\}$. Direct addition gives $D + D = [0, 12]_{\mathbb{Z}}$. Therefore

$$(26 + D) + (26 + D) = [52, 64]_{\mathbb{Z}}.$$

Also $26 + [39, 50]_{\mathbb{Z}} = [65, 76]_{\mathbb{Z}}$ and $77 = 27 + 50$. Thus every element of $[52, 77]_{\mathbb{Z}}$ is decomposable. For $n \geq 78$,

$$n = 26 + (n - 26), \quad n - 26 \geq 52,$$

so every remaining tail element is decomposable too. The minimal generating set is exactly A , all of whose elements are below $F = 51$. Thus S is a leaf.

The gap set is

$$G(S) = [1, 25]_{\mathbb{Z}} \cup \{28, 30, 33, 34, 35, 36, 37, 38, 51\}. \quad (3.3)$$

Every gap in the second set is pseudo-Frobenius, since adding a positive element gives at least $28 + 26 > 51$. For $13 \leq p \leq 18$, adding one of $26, 27, 29, 31, 32$ gives an element of $[39, 50]_{\mathbb{Z}}$, and adding an element of $[39, 50]_{\mathbb{Z}}$ gives at least 52. For $p = 21$, the five smaller-generator sums are 47, 48, 50, 52, 53; for $p = 23$ they are 49, 50, 52, 54, 55. Thus these eight low gaps are pseudo-Frobenius as well.

The remaining seventeen gaps each fail the defining condition, as the witnesses in table 1 show. This proves (3.2) and gives $t = 8 + 9 = 17$. Finally, $34 = 4 \cdot 8 + 2$ gives $\beta(34) = 16$. $\square$

Gap p	Element a	Gap $p + a$	Gap p	Element a	Gap $p + a$
1	27	28	10	26	36
2	26	28	11	26	37
3	27	30	12	26	38
4	26	30	19	32	51
5	29	34	20	31	51
6	27	33	22	29	51
7	26	33	24	27	51
8	26	34	25	26	51
9	26	35			

Table 1: Certificates that the gaps not listed in (3.2) are not pseudo-Frobenius. Every witness a is a minimal generator.

This is a complete finite disproof of the conjecture. In particular, neither the families below nor any asymptotic reasoning is required to establish the contradiction. The purpose of the remaining sections is to explain its structure and identify the correct scale of the extremal problem.

4 Reducing depth-two leaves to finite sets

The counterexample has conductor exactly twice its multiplicity. At this depth, additive closure is automatic, and the nontrivial conditions become finite sumset and translation conditions.

Proposition 4.1. *Let $M \geq 1$, let $m = M + 1$, and let $B \subseteq [0, M]_{\mathbb{Z}}$ satisfy $0 \in B$ and $M \notin B$. Define*

$$S_B = \{0\} \cup (m + B) \cup [2m, \infty)_{\mathbb{Z}}. \quad (4.1)$$

Then S_B is a numerical semigroup with multiplicity m , conductor $2m$, and Frobenius number $2M + 1$. It is a leaf if and only if

$$[0, M]_{\mathbb{Z}} \subseteq B + B. \quad (4.2)$$

For

$$T(B) = \{p \in B \setminus \{0\} : (p + B) \cap [0, M]_{\mathbb{Z}} \subseteq B\}, \quad (4.3)$$

its invariants and pseudo-Frobenius set satisfy

$$g(S_B) = 2M + 1 - |B|, \quad (4.4)$$

$$\text{PF}(S_B) = T(B) \dot{\cup} (m + ([0, M]_{\mathbb{Z}} \setminus B)), \quad (4.5)$$

$$t(S_B) = M + 1 - |B| + |T(B)|. \quad (4.6)$$

When (4.2) holds, the minimal generating set is exactly $m + B$.

Proof. Every positive element is at least m , and every integer at least $2m$ belongs to S_B . Since $m + M = 2m - 1$ is absent, the asserted conductor and Frobenius number are exact. All elements of $m + B$ are less than $2m$, hence primitive.

It remains to examine possible primitive elements in $[2m, 3m - 1]_{\mathbb{Z}}$. A decomposition there cannot use a summand at least $2m$, because the other positive summand would bring the total to at least $3m$. Consequently, $2m + j$ is decomposable exactly when $j = b + b'$ for some $b, b' \in B$. This proves the leaf criterion. Above this interval, $n = m + (n - m)$ is always a decomposition inside the tail.

The positive elements below the conductor are exactly the $|B|$ elements of $m + B$, proving (4.4). Every high gap $m + j$, where $j \in [0, M]_{\mathbb{Z}} \setminus B$, is pseudo-Frobenius: adding m already reaches the tail. A low gap $p \in [1, M]_{\mathbb{Z}}$ can be pseudo-Frobenius only if $p + m \in S_B$, that is, only if $p \in B$. The remaining conditions are

$$p + (m + b) \in S_B \quad (b \in B).$$

For $p + b > M$ they hold automatically, and for $p + b \leq M$ they say exactly $p + b \in B$. This is (4.3). Low and high gaps are disjoint, so taking cardinalities proves the type formula. $\square$

The sumset condition explains the design problem. One wants a relatively small set that covers an initial interval by two-term sums, together with a long interval near the top that is stable under many small positive translations. The former prevents tail generators; the latter creates many low pseudo-Frobenius numbers. The two demands can be separated cleanly.

5 Restricted additive bases produce leaves

Definition 5.1. A *restricted additive basis of order two and maximum d* is a set $D \subseteq [0, d]_{\mathbb{Z}}$ such that

$$D + D = [0, 2d]_{\mathbb{Z}}. \quad (5.1)$$

We assume $d \geq 1$ and write $k = |D|$.

This is the restricted two-summand basis condition studied in the restricted postage-stamp problem; see Kohonen [2] for that terminology and computational setting. We need no extremal theorem from that literature. Equation (5.1) forces $0, 1, d \in D$: representing 0 and $2d$ forces the endpoints, and representing 1 forces 1.

Theorem 5.2 (Basis-to-leaf construction). *Let D satisfy (5.1), and let $\delta \geq 0$ be an integer. Put*

$$\begin{aligned} M &= 4d + 1 + \delta, & m &= M + 1, \\ H &= [2d + 1, M - 1]_{\mathbb{Z}}, & B &= D \cup H, \\ S(D, \delta) &= \{0\} \cup (m + B) \cup [2m, \infty)_{\mathbb{Z}}. \end{aligned} \quad (5.2)$$

Then $S(D, \delta)$ is a leaf, its minimal generators are precisely $m + B$, and

$$m = 4d + 2 + \delta, \quad F = 8d + 3 + 2\delta, \quad c = 8d + 4 + 2\delta, \quad (5.3)$$

$$e = k + 2d + \delta, \quad g = 6d + 3 - k + \delta, \quad t = 4d + 3 - 2k. \quad (5.4)$$

Its pseudo-Frobenius numbers are given explicitly by

$$\begin{aligned} \text{PF}(S(D, \delta)) &= ([M - 2d, M - 1]_{\mathbb{Z}} \setminus \{M - a : a \in D \setminus \{0\}\}) \\ &\quad \dot{\cup} (m + ([0, 2d]_{\mathbb{Z}} \setminus D)) \dot{\cup} \{2M + 1\}. \end{aligned} \quad (5.5)$$

In particular, increasing δ increases the genus by the same amount and leaves the type unchanged.

Proof. The sets D and H are disjoint, $0 \in B$, and $M \notin B$. For the leaf condition, $D + D$ covers $[0, 2d]_{\mathbb{Z}}$, $0 + H$ covers $[2d + 1, M - 1]_{\mathbb{Z}}$, and

$$M = 1 + (M - 1) \in D + H.$$

Thus $B + B$ covers $[0, M]_{\mathbb{Z}}$, so theorem 4.1 proves the leaf property and identifies the generators.

The interval H has $2d + \delta$ elements, whence $|B| = k + 2d + \delta$. This gives the formulas for e and g as well as (5.3). For the type it remains to determine $T(B)$.

First, if $p \in D \setminus \{0\}$, then

$$d + 1 \leq p + d \leq 2d.$$

This interval is disjoint from B , so such p cannot belong to $T(B)$. Second, if $p \in H$ and $p \leq M - 2d - 1$, then $M - p \in H$. The sum $p + (M - p) = M$ is outside B , so this p also fails the condition.

It remains to consider $p \in [M - 2d, M - 1]_{\mathbb{Z}}$. This entire interval lies in H because $M \geq 4d + 1$. For $h \in H$ we have $p + h > M$. For $a \in D$, a sum $p + a \leq M$ is already above $2d$, and hence it belongs to B unless it equals the single missing value M . Therefore

$$T(B) = [M - 2d, M - 1]_{\mathbb{Z}} \setminus \{M - a : a \in D \setminus \{0\}\}. \quad (5.6)$$

The interval has $2d$ elements, and the excluded $k - 1$ values are distinct and lie in it. Thus

$$|T(B)| = 2d - k + 1.$$

The high missing indices are exactly $([0, 2d]_{\mathbb{Z}} \setminus D) \cup \{M\}$, so there are $2d + 2 - k$ high pseudo-Frobenius numbers. Their sum with $|T(B)|$ is $4d + 3 - 2k$, and (5.5) follows from (4.5). $\square$

Corollary 5.3. *Every semigroup from theorem 5.2 satisfies*

$$F + t = 2g, \quad \frac{2g}{3} - t = \frac{4k - 3 + 2\delta}{3}. \quad (5.7)$$

At $\delta = 0$ one also has $2t - g = 2d + 3 - 3k$.

Proof. Substitute (5.3) and (5.4). $\square$

The first identity says that these examples attain the general inequality $F + t \leq 2g$, proved below. The second says that the departure from $2g/3$ is controlled by the basis size, not directly by its maximum. Sparse bases are therefore the natural way to approach the limiting ratio.

6 Two explicit infinite families

6.1 An odd-progression family containing the counterexample

For an integer $q \geq 1$, let

$$d = 2q, \quad D_q = \{0, 2q\} \cup \{1, 3, 5, \dots, 2q - 1\}. \quad (6.1)$$

Then $|D_q| = q + 2$ and $D_q + D_q = [0, 4q]_{\mathbb{Z}}$. Indeed, sums of two odd members give all even integers from 2 through $4q - 2$. The even endpoints come from $0 + 0$ and $2q + 2q$. Adding 0 or $2q$ to the odd members gives all odd integers in the required range.

Corollary 6.1. *The semigroup $S_q = S(D_q, 0)$ is a leaf with*

$$m = 8q + 2, \quad F = 16q + 3, \quad e = 5q + 2, \quad g = 11q + 1, \quad t = 6q - 1. \quad (6.2)$$

For every $q \geq 3$ it contradicts (1.1).

Proof. Apply theorem 5.2. The excess over half the genus is

$$2t - g = 2(6q - 1) - (11q + 1) = q - 3.$$

For $q \geq 3$, this is nonnegative; now use (1.2). $\square$

At $q = 3$, $D_q = \{0, 1, 3, 5, 6\}$, $M = 25$, and $H = [13, 24]_{\mathbb{Z}}$, recovering (3.1). Thus the genus-34 example is not isolated. Its successors increase the discrepancy without bound, and t/g tends to $6/11 > 1/2$.

q	g	t	$\beta(g)$	$t - \beta(g)$	$2t - g$
1	12	5	5	0	-2
2	23	11	11	0	-1
3	34	17	16	1	0
4	45	23	21	2	1
5	56	29	27	2	2
6	67	35	33	2	3
7	78	41	38	3	4
8	89	47	43	4	5

Table 2: The first odd-progression examples. The entries are direct evaluations of proved formulas, also checked by the supplied program.

6.2 A sparse square-grid family

A different elementary basis improves the limiting ratio to $2/3$. For $r \geq 1$, put $d = r^2$ and define

$$D_r = [0, r-1]_{\mathbb{Z}} \cup \{0, r, 2r, \dots, r^2\} \cup [r^2 - r + 1, r^2]_{\mathbb{Z}}. \quad (6.3)$$

The set contains $3r - 1$ elements. It is symmetric under $a \mapsto r^2 - a$. For $0 \leq n \leq r^2$, Euclidean division gives

$$n = qr + s, \quad 0 \leq s < r,$$

with both qr and s in D_r . Reflecting this representation about r^2 represents every $n \in [r^2, 2r^2]_{\mathbb{Z}}$ as well. Consequently, $D_r + D_r = [0, 2r^2]_{\mathbb{Z}}$.

Corollary 6.2. *For $S_r = S(D_r, 0)$, write $g_r = g(S_r)$ and $t_r = t(S_r)$. Then*

$$g_r = 6r^2 - 3r + 4, \quad t_r = 4r^2 - 6r + 5. \quad (6.4)$$

These are leaves, and

$$t_r = \frac{2}{3}g_r - 4r + \frac{7}{3}, \quad \lim_{r \rightarrow \infty} \frac{t_r}{g_r} = \frac{2}{3}. \quad (6.5)$$

They contradict the original conjecture for every $r \geq 4$.

Proof. The formulas follow by substituting $d = r^2$ and $k = 3r - 1$ into theorem 5.2. Moreover,

$$2t_r - g_r = 2r^2 - 9r + 6,$$

which is positive at $r = 4$ and strictly increasing for integer $r \geq 4$. The contradiction again follows from (1.2). $\square$

6.3 Why square-root defects arise in this construction

There are at most $k(k+1)/2$ distinct unordered two-term sums from a set of size k , allowing equal summands. A restricted basis therefore satisfies

$$\frac{k(k+1)}{2} \geq 2d + 1, \quad k \geq \frac{\sqrt{16d+9} - 1}{2}.$$

r	$ D_r $	g_r	t_r	$\beta(g_r)$	$t_r - \beta(g_r)$
1	2	7	3	3	0
2	5	22	9	10	-1
3	8	49	23	23	0
4	11	88	45	43	2
5	14	139	75	69	6
6	17	202	113	100	13
8	23	364	213	181	32
10	29	574	345	286	59
20	59	2344	1485	1171	314
30	89	5314	3425	2656	769

Table 3: Sparse-grid leaves. The quadratic growth of genus and type, versus the linear size of the additive basis, yields the limiting ratio $2/3$.

Thus its size is necessarily at least of square-root order in d . The bases (6.3) have size $3\sqrt{d} - 1$, which is sufficient for our asymptotic purpose even without any claim of optimality. By (5.7), these bounds translate directly into square-root defects from $2g/3$ along the unpadding family.

This counting observation applies only to this construction. The next section gives a separate argument showing that a square-root defect is unavoidable for *every* leaf, whether or not it has conductor $2m$ or comes from a restricted basis.

7 Universal upper bounds for the type of a leaf

7.1 The reflection inequality

The inequality in the following lemma is standard; it also appears as Proposition 3.1 of the target paper [1]. We include a direct proof so that all later bounds can be checked from the definitions.

Lemma 7.1. *Every nontrivial numerical semigroup satisfies*

$$F + t \leq 2g. \quad (7.1)$$

Proof. Let $N = S \cap [1, F]_{\mathbb{Z}}$. Its cardinality is $F - g$. The sets N and $\text{PF}(S) \setminus \{F\}$ are disjoint. Reflection $x \mapsto F - x$ sends their union injectively into $G(S) \setminus \{F\}$.

For $x \in N$, membership $F - x \in S$ would imply $F \in S$ by closure. For $x \in \text{PF}(S) \setminus \{F\}$, membership of the positive number $F - x$ in S would contradict the pseudo-Frobenius condition. All reflected values are positive and smaller than F . Comparing cardinalities gives

$$(F - g) + (t - 1) \leq g - 1,$$

which is equivalent to (7.1). □

Since $t \geq 1$, this also gives $F \leq 2g - 1$. Thus there are only finitely many numerical semigroups of any fixed positive genus: their gap sets are subsets of the finite interval $[1, 2g - 1]_{\mathbb{Z}}$. This justifies the finiteness asserted in the definition of $L(g)$.

7.2 Apéry sets and nonmaximal elements

Definition 7.2. For the multiplicity m , the *Apéry set* is

$$W = \text{Ap}(S; m) = \{s \in S : s - m \notin S\}.$$

It contains the least element of S in each residue class modulo m , hence has m elements and contains 0. On W use the partial order

$$u \preceq_S v \iff v - u \in S.$$

The numerically largest Apéry element is $F + m$. Indeed, it belongs to S while subtracting m gives the gap F ; conversely, $w - m \notin S$ prevents an Apéry element w from exceeding $F + m$.

Lemma 7.3. *The map $p \mapsto p + m$ is a bijection from $\text{PF}(S)$ to the maximal elements of $(W, \preceq_S)$. In particular, precisely t elements of W are maximal, and none of them is 0.*

Proof. For $p \in \text{PF}(S)$, $p + m \in S$ and $p \notin S$, so $p + m \in W$. If $w \in W$ strictly dominates $p + m$, write $w = p + m + s$ with positive $s \in S$. Then $w - m = p + s \in S$, a contradiction.

Conversely, let w be maximal in W . It is nonzero because 0 is below every element of W and $m \geq 2$. Every nonzero Apéry element exceeds m , so $p = w - m$ is a positive gap. If $p + s \notin S$ for some positive $s \in S$, then $w + s \in W$, contradicting maximality. Thus $p \in \text{PF}(S)$. $\square$

Write

$$a = m - 1 - t \tag{7.2}$$

for the number of nonzero, nonmaximal Apéry elements.

Lemma 7.4. *For every leaf, $a \geq 1$ and $F \geq 2m - 1$. Consequently, with $h = 2g - 3t$, one has*

$$h \geq 2a + 1 \geq 3, \quad t \leq \left\lfloor \frac{2g - 3}{3} \right\rfloor. \tag{7.3}$$

Proof. The Apéry element $F + m$ lies above F and therefore is not primitive in a leaf. Write $F + m = u + v$ with $u, v \in S \setminus \{0\}$. Both u and v belong to W : if $u - m \in S$, for example, then $F = (u - m) + v \in S$, a contradiction. Both are nonmaximal since $u \prec_S F + m$ and $v \prec_S F + m$. They may coincide, but at least one nonzero nonmaximal element exists. Hence $a \geq 1$.

If $F < 2m - 1$, the number $2m - 1$ belongs to S and is smaller than the sum of any two positive elements. It is therefore a primitive element above F , contrary to the leaf condition. Thus $F \geq 2m - 1$.

Combining this with theorem 7.1 and $m = t + a + 1$ gives

$$2g - 3t \geq F - 2t \geq 2m - 1 - 2t = 2a + 1.$$

The final bound follows from $a \geq 1$. $\square$

7.3 Counting decompositions above the Frobenius number

The preceding linear bound identifies the candidate leading constant $2/3$, but it does not yet rule out a bounded defect from that line. A second count forces many nonmaximal Apéry elements, which creates the missing square-root term.

Lemma 7.5. *For any leaf, with a as in (7.2),*

$$\frac{a(a+1)}{2} \geq 2t + a - g. \quad (7.4)$$

Proof. Consider $w \in W$ with $w > F$. Since S is a leaf, $w = u + v$ for positive $u, v \in S$. The argument used in theorem 7.4 works for this w as well: if $u - m \in S$, then $w - m = (u - m) + v \in S$, contradicting $w \in W$. Thus both summands are in W , and both are nonmaximal because each is strictly below w in the semigroup order.

There are only $a(a+1)/2$ unordered pairs, with repetition, of nonzero nonmaximal Apéry elements. Their sums can therefore cover at most that many distinct values of w . Hence

$$|\{w \in W : w > F\}| \leq \frac{a(a+1)}{2}.$$

On the other hand, the number of nonzero Apéry elements at most F is at most the number $n = F - g$ of positive semigroup elements at most F . Consequently,

$$|\{w \in W : w > F\}| \geq m - 1 - n.$$

By $F + t \leq 2g$ we have $n = F - g \leq g - t$, so

$$m - 1 - n \geq t + a - (g - t) = 2t + a - g.$$

Combining these estimates proves (7.4). □

Theorem 7.6 (Square-root upper bound). *Every leaf of genus g and type t satisfies*

$$(2g - 3t)^2 \geq 4t - 3, \quad (7.5)$$

and, more explicitly,

$$t \leq \left\lfloor \frac{6g + 2 - \sqrt{24g - 23}}{9} \right\rfloor. \quad (7.6)$$

Proof. Set $h = 2g - 3t$. Rearranging (7.4) gives

$$a^2 - a \geq 4t - 2g = t - h, \quad \text{or} \quad t \leq a^2 - a + h.$$

From theorem 7.4, $1 \leq a \leq (h - 1)/2$. The function $x^2 - x$ is increasing on $[1, \infty)$, so

$$t \leq \left(\frac{h-1}{2}\right)^2 - \frac{h-1}{2} + h = \frac{h^2 + 3}{4}.$$

This proves (7.5).

To obtain the displayed bound on t , substitute $t = (2g - h)/3$ to get

$$3h^2 + 4h + 9 - 8g \geq 0.$$

Since $h \geq 3$, this implies

$$h \geq \frac{\sqrt{24g - 23} - 2}{3}.$$

Now use $t = (2g - h)/3$ and the integrality of t . □

The decisive count is of *nonmaximal* Apéry elements, not of all minimal generators. Every high Apéry element needs a decomposition, and its summands must be nonmaximal Apéry elements. A small supply of such elements generates only quadratically many sums. Near the extremal line, there are linearly many high elements to cover, so the supply must grow at least like a square root. The inequalities above make that observation quantitative without imposing a special form on S .

8 The maximum type at every large genus

The unpadded grid family proves a lower bound only at the genera $g_r = 6r^2 - 3r + 4$. To determine an ordinary limit rather than just a limit superior, one must fill the gaps between these genera. The parameter δ in theorem 5.2 does exactly that.

Theorem 8.1. *For every integer $g \geq 7$, there is a leaf of genus g . Moreover,*

$$L(g) \geq \frac{2g}{3} - \sqrt{24g - 87} - 2. \quad (8.1)$$

Together with theorem 7.6, this yields

$$L(g) = \frac{2}{3}g - \Theta(\sqrt{g}), \quad \lim_{g \rightarrow \infty} \frac{L(g)}{g} = \frac{2}{3}. \quad (8.2)$$

Proof. The grid basis with $r = 1$ has genus $g_1 = 7$ and type 3; arbitrary $\delta \geq 0$ already proves existence at every genus at least 7. For the stronger lower bound, choose the largest $r \geq 1$ such that $g_r \leq g$, equivalently

$$r = \left\lfloor \frac{3 + \sqrt{24g - 87}}{12} \right\rfloor. \quad (8.3)$$

Set $\delta = g - g_r$. Since

$$g_{r+1} - g_r = 12r + 3,$$

we have $0 \leq \delta \leq 12r + 2$. By theorem 5.2, $S(D_r, \delta)$ has exactly genus g and still has type t_r . Using (6.5),

$$\begin{aligned} L(g) &\geq t_r = \frac{2}{3}(g - \delta) - 4r + \frac{7}{3} \\ &\geq \frac{2}{3}g - 12r + 1. \end{aligned}$$

Equation (8.3) implies $12r \leq 3 + \sqrt{24g - 87}$, proving (8.1).

Let $E(g) = 2g/3 - L(g)$. The lower and upper bounds together imply

$$\frac{\sqrt{24g - 23} - 2}{9} \leq E(g) \leq \sqrt{24g - 87} + 2, \quad g \geq 7. \quad (8.4)$$

Both outer expressions have positive square-root leading terms. Hence $E(g) = \Theta(\sqrt{g})$, and division by g proves the asserted limit. $\square$

Here $\Theta(\sqrt{g})$ has its usual two-sided meaning: there exist positive constants c, C and a threshold g_0 such that $c\sqrt{g} \leq 2g/3 - L(g) \leq C\sqrt{g}$ for every integer $g \geq g_0$. It does not assert the existence of a limiting square-root coefficient.

There is also an explicit, though not optimized, point beyond which a counterexample exists at every genus.

Corollary 8.2. *For every integer $g \geq 697$, the construction above gives a leaf with $t > \beta(g)$.*

Proof. Since $g_{11} = 697$, the choice in (8.3) gives $r \geq 11$. For $0 \leq \delta \leq 12r + 2$,

$$\begin{aligned} 2t_r - (g_r + \delta) &\geq (2r^2 - 9r + 6) - (12r + 2) \\ &= 2r^2 - 21r + 4 > 0. \end{aligned}$$

The final polynomial is positive at $r = 11$ and increasing thereafter. Thus $t_r \geq g/2$, and (1.2) gives the contradiction. $\square$

The correction to the original picture is therefore not a sporadic exception or a change of a rounding term. The proposed bound has leading coefficient $1/2$, whereas the actual extremal function has leading coefficient $2/3$, with a negative correction of square-root order.

9 Search, exact verification, and reproducibility

9.1 Discovery was optimization; validation is exact

The initial search used fixed multiplicity m and Frobenius number F . For $1 \leq i \leq F$, binary variables s_i indicated semigroup membership and binary variables p_i marked proposed pseudo-Frobenius numbers. Membership was fixed by

$$s_1 = \cdots = s_{m-1} = 0, \quad s_m = 1, \quad s_F = 0.$$

For indices within $[1, F]_{\mathbb{Z}}$, closure and the pseudo-Frobenius implications were encoded by

$$\begin{aligned} s_i + s_j - s_{i+j} &\leq 1 & (i + j \leq F), \\ p_i + s_i &\leq 1, \\ p_i + s_j - s_{i+j} &\leq 1 & (i + j \leq F). \end{aligned}$$

Only $j \geq m$ is needed in the last constraint, and only $i, j \geq m$ in the closure constraint. Repeated summands have coefficient 2 where appropriate.

To ensure the leaf condition, for each $n \in [F + 1, F + m]_{\mathbb{Z}}$ we introduced binary witnesses z_{ij} for $i + j = n$ with $m \leq i \leq j \leq F$, imposed $z_{ij} \leq s_i, s_j$, and required at least one witness. Such decompositions cannot use a summand greater than F , because then their total with a positive summand would exceed $F + m$.

The maximized expression was

$$2 \sum_{i=1}^F p_i + \sum_{i=1}^F s_i - F.$$

For any fixed membership vector, maximizing the p_i marks every valid pseudo-Frobenius gap, so this expression becomes $2t - g$. Positive or zero values are a convenient sufficient test for violating (1.1). This objective need not find every possible parity-dependent violation.

The optional implementation uses SciPy's mixed-integer linear programming interface [3]. An initial run at $(m, F) = (36, 71)$ returned a leaf with $(g, t) = (48, 24)$, exceeding the bound 23. Examining its low elements led to the restricted-basis construction, and then to the smaller $(34, 17)$ example. The historical candidates are saved in `data/discovery_search.jsonl`.

No floating-point optimality claim is used in the proofs. Search results are merely proposals: every candidate used in the article is reconstructed and checked with integer arithmetic. A time-limited optimizer may return a different candidate, or no candidate, on another machine without affecting any theorem in this note.

9.2 Two independent descriptions of the main example

The principal verifier first starts with the finite-tail description $S = \{0\} \cup A \cup [c, \infty)\mathbb{Z}$. It checks closure below c , finds the exact gap set, tests the pseudo-Frobenius condition against all positive elements below c , and searches for primitive elements through $F + m$. Integer bitsets make these tests fast while preserving exactness.

A second method starts only from the listed generators. Form a directed graph on residues modulo m ; a generator a gives an edge of weight a from residue r to residue $r + a$. Shortest-path distances from 0 are exactly the least representable nonnegative integers in each residue, namely the Apéry set. The supplied Dijkstra implementation uses integer weights. Unreachable residues would reveal that the generators do not have greatest common divisor 1.

For (3.1), the Apéry vector, in residue order modulo 26, is

$$(w_0, \dots, w_{25}) = (0, 27, 54, 29, 56, 31, 32, 59, 60, 61, 62, 63, 64, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 77). \quad (9.1)$$

Membership is then characterized by

$$n \in S \iff n \geq w_{n \bmod m} \quad (n \geq 0).$$

The maximum entry gives $F = 77 - 26 = 51$. Counting missing elements residue by residue gives the independent genus formula

$$g = \sum_{i=0}^{m-1} \frac{w_i - i}{m} = 34.$$

Maximal Apéry elements recover the same pseudo-Frobenius set by theorem 7.3. The code also independently recomputes all primitive elements from this membership test. Thus the tail description and the generator description are checked against one another, rather than merely re-evaluating the same closed formulas twice.

9.3 Scope of the finite checks

The default run completed the checks in table 4; its results are saved in `data/verification_report.json`.

Check	Count or range
Odd-progression and padded square-grid instances	220
Restricted bases with maximum $1 \leq d \leq 12$	384
Those bases checked at $\delta = 0, 1, 7$	1,152
Distinct examples reconstructed independently by Dijkstra	26
Exhaustive tree levels checked	genus 1–20
Semigroups in those levels	93,141
Leaves in those levels	34,001

Table 4: Exact finite verification completed for this archive. The infinite claims are established by the written proofs, not inferred from these counts.

For each constructed example, the program checks closure, the leaf condition, the precise generator set, the genus, the entire pseudo-Frobenius set, the type formula, and the universal linear and

quadratic inequalities. The exhaustive tree traversal generates children by deleting every primitive above the Frobenius number. Each child has a unique parent, so starting at genus one and traversing every child covers every semigroup at the next genus without duplicates. The original bound holds on all leaves through genus 20 in this run.

For orientation, the independently computed small-genus maxima are

g	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$L(g)$	3	3	3	4	5	5	5	6	7	7	7	8	9	9.

These small values explain why limited enumeration does not distinguish the proposed half-genus scale from a larger eventual scale.

The checks are ordinary executable integer computations, not proofs checked in a foundational proof assistant. Their purpose is to catch arithmetic, implementation, and transcription errors, especially in the finite certificates. They do not certify priority, global minimality of the counterexample, or optimality of a basis or semigroup beyond the explicitly enumerated ranges.

10 What is settled and what remains

The original conjecture is settled negatively by theorem 3.1. The same construction supplies infinitely many counterexamples by theorem 6.1, and counterexamples at every genus at least 697 by theorem 8.2. The maximum leaf type has the exact leading coefficient $2/3$, and the magnitude of its deficit from that line is exactly square-root order by theorem 8.1.

Several more precise questions remain outside those conclusions. First, the exact values of $L(g)$ beyond the small enumerated range are not determined here. A constructed type gives a lower bound, not necessarily the maximum. Second, the ratio

$$\frac{2g/3 - L(g)}{\sqrt{g}}$$

is bounded above and away from zero for large g , but the present argument does not show that it converges. Along the unpadded square-grid sequence, our examples have deficit $4r - 7/3$, whereas padding gives a larger uniform constant. Better restricted bases, denser choices of their maxima, or constructions of a different shape could improve the lower-bound constant.

Third, the universal upper bound deliberately forgets information about which pairs of nonmaximal Apéry elements may have equal sums and which sums can occur in the relevant residue classes. Retaining that structure could strengthen the square-root coefficient. The elementary estimate $a(a+1)/2$ is enough for the order of growth but is not asserted to be sharp for this problem.

Finally, a genus-minimal counterexample would require either a complete independent search below genus 34 or a structural exclusion theorem. The example here is at the first genus after the source paper's reported range, but our independent search stops at 20. No stronger minimality statement is needed for the disproof or the asymptotic theorem.

The reusable mathematical mechanism is the combination of two finite objects: a restricted additive basis that guarantees enough decompositions, and an Apéry poset whose nonmaximal elements account for the cost of those decompositions. The construction and the upper bound meet at the same square-root scale, even though they arise from different counts.

A Additional certificate for the conductor interval

To make the generator claim independently inspectable, here are explicit decompositions of the entire possible-tail-generator interval $[52, 77]_{\mathbb{Z}}$. The compact proof in section 3 groups these into three blocks; the table lists the individual witnesses.

Lower half	Upper half
$52 = 26 + 26$	$65 = 26 + 39$
$53 = 26 + 27$	$66 = 26 + 40$
$54 = 27 + 27$	$67 = 26 + 41$
$55 = 26 + 29$	$68 = 26 + 42$
$56 = 27 + 29$	$69 = 26 + 43$
$57 = 26 + 31$	$70 = 26 + 44$
$58 = 26 + 32$	$71 = 26 + 45$
$59 = 27 + 32$	$72 = 26 + 46$
$60 = 29 + 31$	$73 = 26 + 47$
$61 = 29 + 32$	$74 = 26 + 48$
$62 = 31 + 31$	$75 = 26 + 49$
$63 = 31 + 32$	$76 = 26 + 50$
$64 = 32 + 32$	$77 = 27 + 50$

All summands occur in the proposed generating set. Above 77, subtracting 26 stays in the full tail, completing the argument for every larger integer. The machine-readable certificate in `data/counterexample.json` contains these decompositions, the witnesses in table 1, the complete gap set, the pseudo-Frobenius set, and the residue-ordered Apéry vector.

B Reproducing the random draw and the computations

The area-selection method was `random.Random(secrets.randbits(128)).randrange(24)`. The sampled seed was

84421886621299946913970799425001841153.

The returned zero-based index was 9, selecting semigroup theory. The complete ordered list of areas is in `data/random_selection.json`. Replaying the seed verifies the recorded index; it does not require fresh randomness.

From the extracted archive directory, run

```
python code/verify.py --out data
```

The verifier requires only the Python standard library and uses integer arithmetic. Python 3.10 or later supports its syntax; the saved run used Python 3.13.5. Verification failures raise exceptions through explicit checks, so running Python with optimization does not disable them. The optional argument `-tree-genus 24` extends the exhaustive traversal but is not necessary for the default report.

The optional numerical discovery program is separate:

```
python -m pip install -r requirements-search.txt
python code/search.py --multiplicity 36 --frobenius 71 --seconds 15
```

The original environment used NumPy 2.3.5 and SciPy 1.17.0. Candidate validity is checked exactly by the verifier even when a solver stops at its time limit. Solver optimality is not part of the certificate.

To compile the article, run

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Alternatively, run `pdflatex article.tex` twice. All references are included directly in the source, so no bibliography database or external image is required. The archive includes both the editable source and the compiled PDF, together with a README, verification data, optional search requirements, and SHA-256 checksums.

References

- [1] Jonathan Chappelon, Jorge L. Ramírez Alfonsín, and Dumitru I. Stamate, *Numerical semigroup tree: type-representation*, arXiv:2507.15006v2, revised 22 November 2025. <https://arxiv.org/abs/2507.15006v2>. Conjecture 4.4 is the target of this note.
- [2] Jukka Kohonen, *Early Pruning in the Restricted Postage Stamp Problem*, arXiv:1503.03416, 2015. <https://arxiv.org/abs/1503.03416>.
- [3] SciPy developers, *scipy.optimize.milp — Mixed-integer linear programming*, SciPy reference documentation, accessed 20 September 2026. <https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.milp.html>.